Practical 7

Aim: Write shell script for the following -

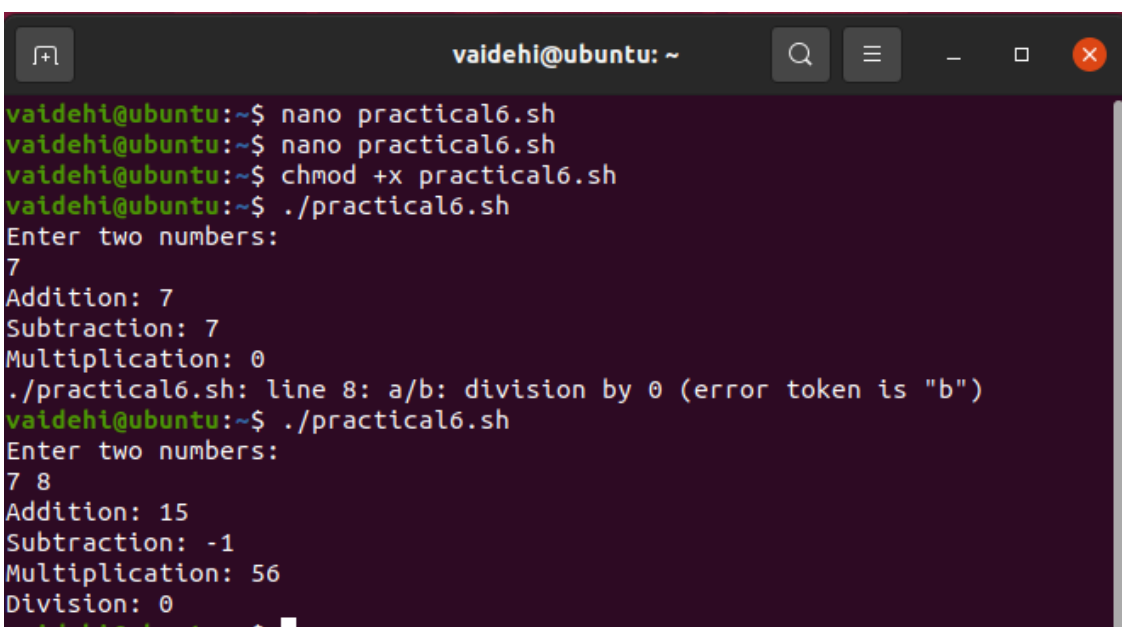
Theory:

Explain File and Directory System in the operating system and its use.The file and directory system in an operating system is used to organize and manage data stored on a computer. A file is a collection of related data, while a directory is a folder that stores files and other directories in a structured way. The system provides a hierarchical structure, making it easy to locate and access data. It also supports operations like creating, deleting, reading, and writing files, along with security through access permissions. Overall, it helps in efficient storage, organization, and retrieval of data.

- Write a shell script to perform arithmetic operations.

```bash
echo "Enter two numbers:"
read a b

echo "Addition: $((a+b))"
echo "Subtraction: $((a-b))"
echo "Multiplication: $((a*b))"
echo "Division: $((a/b))"
```



- Write a shell script to calculate Simple Interest and Compound Interest calculation.

```bash
echo "Enter Principal, Rate, Time:"
read p r t

si=$((p*r*t/100))
ci=$(echo "$p*(1+$r/100)^$t" | bc)

echo "Simple Interest: $si"
echo "Compound Interest: $ci"
```

```
vaidehi@ubuntu:~$ nano practical6.sh


Use "fg" to return to nano.

[2]+  Stopped                 nano practical6.sh
vaidehi@ubuntu:~$ nano practical6.sh
vaidehi@ubuntu:~$ chmod +x practical6.sh
vaidehi@ubuntu:~$
vaidehi@ubuntu:~$ ./practical6.sh
Enter Principal, Rate, Time:
100000 2 2
Simple Interest: 4000
Compound Interest: 100000
vaidehi@ubuntu:~$
```

- Write a shell script to determine if a given year is a leap year or Not.

```bash
echo "Enter year:"
read year

if (( (year%4==0 && year%100!=0) || (year%400==0) ))
then
    echo "Leap Year"
else
    echo "Not Leap Year"
fi
```

```
vaidehi@ubuntu:~$ nano practical6.sh
vaidehi@ubuntu:~$ nano practical6.sh
vaidehi@ubuntu:~$ chmod +x practical6.sh
vaidehi@ubuntu:~$ ./practical6.sh
Enter year:
2006
Not Leap Year
vaidehi@ubuntu:~$
```

- Write a shell script to compare two strings.

```bash
echo "Enter first string:"
read str1
echo "Enter second string:"
read str2

if [ "$str1" = "$str2" ]
then
    echo "Strings are equal"
else
    echo "Strings are not equal"
fi
```

```
vaidehi@ubuntu:~$ nano practical6.sh
vaidehi@ubuntu:~$ chmod +x practical6.sh
vaidehi@ubuntu:~$ ./practical6.sh
Enter first string:
Vaidehi
Enter second string:
Deshpande
Strings are not equal
vaidehi@ubuntu:~$
```

- Write a shell script to implement a menu-driven calculator using a case statement.

```bash
echo "1.Add 2.Subtract 3.Multiply 4.Divide"
echo "Enter choice:"
read ch

echo "Enter two numbers:"
read a b

case $ch in
1) echo "Result: $((a+b))" ;;
2) echo "Result: $((a-b))" ;;
3) echo "Result: $((a*b))" ;;
4) echo "Result: $((a/b))" ;;
*) echo "Invalid choice" ;;
esac
```

```
vaidehi@ubuntu:~$ nano practical6.sh
vaidehi@ubuntu:~$ chmod +x practical6.sh
vaidehi@ubuntu:~$ ./practical6.sh
1.Add 2.Subtract 3.Multiply 4.Divide
Enter choice:
1
Enter two numbers:
8  9
Result: 17
vaidehi@ubuntu:~$ ./practical6.sh
1.Add 2.Subtract 3.Multiply 4.Divide
Enter choice:
4
Enter two numbers:
20 4
Result: 5
```

- Write a shell script for Employee Pay calculation.

```bash
echo "Enter basic salary:"
read basic

da=$((basic*40/100))
hra=$((basic*20/100))
gross=$((basic+da+hra))

echo "DA: $da"
echo "HRA: $hra"
echo "Gross Salary: $gross"
```

```
vaidehi@ubuntu:~$ nano practical6.sh
vaidehi@ubuntu:~$ chmod +x practical6.sh
vaidehi@ubuntu:~$ ./practical6.sh
Enter basic salary:
10000000
DA: 4000000
HRA: 2000000
Gross Salary: 16000000
vaidehi@ubuntu:~$
```

- Write a shell script to perform operations on directory like:
  - display name of current directory;
  - display list of directory
  - contents;
  - create another directory,
  - write contents on that and
  - copy it to a suitable location in your home directory; etc

```bash
echo "Current Directory:"
pwd

echo "List of files:"
ls

echo "Enter new directory name:"
read d

mkdir $d
cd $d

echo "Hello this is file" > file1.txt

cp file1.txt ~
```

```
vaidehi@ubuntu:~$ nano practical7.sh
vaidehi@ubuntu:~$ chmod +x practical7.sh
vaidehi@ubuntu:~$ ./practical7.sh
Current Directory:
/home/vaidehi
List of files:
2             file_stat.c.save    practical6.sh          syscalls.c
Desktop       file_stat.c.save.1  practical6.sh.save     syscalls.c.save
Documents     fork_exec           practical6.sh.save.1   Templates
Downloads     fork_exec.c         practical6.sh.save.2   test.txt
file1.txt     Music               practical7.sh          Vaidehi
file2.txt     mypatch.patch       Public                 Videos
file_stat     OS_Practical2_Work  snap
file_stat.c   Pictures            syscalls
Enter new directory name:
vaidehi
```

- Write a shell script to search whether an element is present in the list or not.

```bash
echo "Enter elements:"
read -a arr

echo "Enter element to search:"
read key

for i in "${arr[@]}"
do
  if [ $i -eq $key ]
  then
    echo "Found"
    exit
  fi
done

echo "Not Found"
```

```
vaidehi@ubuntu:~$ chmod +x practical7.sh
vaidehi@ubuntu:~$ ./practical7.sh
Enter elements:
9026646
Enter element to search:
6
Not Found
```

- Write a shell script to read and check if the directory / file exists or not, if not make the directory / file.

```bash
echo "Enter file/directory name:"
read name

if [ -e $name ]
then
  echo "Exists"
else
  mkdir $name
  echo "Created"
fi
```

```
vaidehi@ubuntu:~$ chmod +x practical7.sh
vaidehi@ubuntu:~$ ./practical7.sh
Enter elements:
vaidehi
Enter element to search:
file
./practical7.sh: line 10: [: vaidehi: integer expression expected
Not Found
```

Conclusion:

This practical demonstrates how shell scripting can automate tasks like calculations, decision-making, and menu-driven programs. It helps in improving efficiency and reduces manual work in operating systems. We also performed different operations on directory.